PROJECT REPORT

Data Modeling Project using Medallion Architecture and Delta Lake

Celebal Excellence Internship

Name : Lovepreet Kaur

CEI ID: CT_CSI_DE_68

Branch : BTECH CSE

College Name : MMDU

Global Retail Data Modelling Project Report

1. Project Overview

The objective of this project was to design and implement a modern data warehouse for a retail organization using the **Medallion Architecture (Bronze, Silver and Gold layers)** in Databricks. Data from CRM and ERP systems was ingested, cleaned, standardized, integrated, and transformed into a Star Schema to support analytical reporting.

2. Technology Stack

- **Platform:** Databricks Free Edition
- **Language:** PySpark, Spark SQL
- **Storage Format:** Delta Lake
- **Architecture:** Medallion Architecture
- **Data Model:** Star Schema

3. Bronze Layer

The Bronze layer stores the raw source data without any transformation. Each CSV file was loaded into a separate Delta table under the **bronze** schema.

Bronze Tables

Source	**Table**
CRM Customer	bronze.crm_cust_info
CRM Product	bronze.crm_prd_info
CRM Sales	bronze.crm_sales_details
ERP Customer	bronze.erp_cust_az12
ERP Location	bronze.erp_loc_a101

Source	Table
ERP Product Category	bronze.erp_px_cat_g1v2

The purpose of this layer is to preserve the original source data for auditing and reprocessing.

4. Silver Layer

The Silver layer performs data cleansing, validation and standardization before loading the data into the analytical model.

Customer Data (silver.crm_cust_info)

The following transformations were performed:

- Removed duplicate customer records based on cst_id.
- Standardized cst_gndr values to **Male** and **Female**.
- Standardized cst_marital_status values to **Married** and **Single**.
- Preserved valid customer records for further integration.

ERP Customer (silver.erp_cust_az12)

Transformations:

- Standardized GEN values (Male, Female).
- Removed blank and inconsistent gender values.
- Preserved customer birth date (BDATE).

ERP Location (silver.erp_loc_a101)

Transformations:

- Standardized country values.

Examples:

- USA → United States
- US → United States

- DE → Germany
- Blank values → NULL

Product Data (silver.crm_prd_info)

Transformations:

- Standardized product line values into descriptive names.
- Preserved product validity dates (prd_start_dt, prd_end_dt).
- Derived:
    - category_id
    - product_code

The product_code was generated from prd_key to match product identifiers stored in the sales table.

Example:

BI-RB-BK-R93R-44
↓

BK-R93R-44

Product Category (silver.erp_px_cat_g1v2)

The dataset required no data cleaning because:

- No duplicate records
- No missing values
- No invalid categorical values

Sales Data (silver.crm_sales_details)

The following validations were performed:

- Converted integer date columns into Date format.
- Replaced invalid dates with NULL.
- Validated sales calculations.

- Identified negative sales and invalid prices.
- Standardized product keys for joining with Product Dimension.

5. Gold Layer

The Gold layer organizes the cleaned data into a **Star Schema**.

Customer Dimension (gold.dim_customer)

The Customer Dimension was created by integrating:

- silver.crm_cust_info
- silver.erp_cust_az12
- silver.erp_loc_a101

Customer identifiers were standardized before joining:

- Removed NAS prefix from ERP customer IDs.
- Removed hyphens (-) from location customer IDs.

A surrogate key (customer_sk) was generated using row_number().

Important Columns

- customer_sk
- customer_id
- customer_key
- first_name
- last_name
- gender
- marital_status
- birth_date
- country

Product Dimension (gold.dim_product)

The Product Dimension was created by joining:

- silver.crm_prd_info
- silver.erp_px_cat_g1v2

The project derived a custom **product_code** from prd_key to match the sales product identifier.

A surrogate key (product_sk) was generated using row_number().

Important Columns

- product_sk
- product_id
- product_key
- product_code
- product_name
- product_cost
- product_line
- category
- subcategory
- maintenance
- start_date
- end_date

Sales Fact Table (gold.fact_sales)

The Fact table was created from:

- silver.crm_sales_details
- gold.dim_customer
- gold.dim_product

Foreign keys:

- customer_sk
- product_sk

Measures:

- quantity
- price
- sales

Dates:

- order_date
- ship_date
- due_date

Business Key:

- order_number

6. Data Modeling

A **Star Schema** was implemented consisting of:

Fact Table

- fact_sales

Dimension Tables

- dim_customer
- dim_product

The fact table references both dimensions through surrogate keys, enabling efficient analytical queries.

7. Key Transformations

- Standardized customer identifiers across CRM and ERP datasets.
- Generated product_code from prd_key to establish correct joins between product and sales datasets.
- Standardized gender, marital status and country values.
- Removed duplicate customer records.

- Validated and standardized date columns.
- Generated surrogate keys (customer_sk, product_sk) for dimensional modeling.
- Stored all layers as Delta Tables.

8. Project Outcome

The project successfully transformed six raw CRM and ERP datasets into a centralized data warehouse following the Medallion Architecture. The final Gold layer contains a Star Schema with fact_sales, dim_customer, and dim_product, enabling reliable and efficient analytical reporting while maintaining a clear separation between raw, cleansed, and business-ready data.